First High-Resolution Tomographic View of the Circumgalactic Medium

The $z = 0.73$ Mg II - Fe II absorber toward the Sunburst Arc

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ABSTRACT

We present the first tomographic view of the circumgalactic medium (CGM) at high spectral resolution, using the brightest known gravitationally lensed arc, the 'Sunburst Arc'. We study an intervening Mg II – Fe II absorption system at $z \sim 0.73$, associated with an identified galaxy. Our analysis is based on UVES/VLT spectroscopy of eight individual sight lines, which probe a continuous range of projected distances from 3 to 28 kpc from the galaxy.

We detect strong absorption from the cool CGM ($T \sim 10^4$ K) across the lensed arc, with a covering fraction close to unity. The absorption remains strong and saturated out to projected distances of approximately 25 kpc. Despite the high spectral resolution ($R \simeq 30,000$), we do not resolve multiple velocity components in any of the pseudo-spaxels. We model the Mg II and Fe II absorption lines using single-component Voigt profiles and find large Doppler parameters. These broad profiles are likely produced by the combined velocity dispersion of multiple unresolved absorbing components within each pseudo-spaxel.

Comparison with quasar sight lines and simulated pseudo-spaxel observations shows that the recovered column densities are affected by both Mg II saturation and spatial averaging. The increase in the Mg II/Fe II ratio toward larger galactocentric distances along the major axis is qualitatively consistent with an α-enhanced outer CGM, although ionization and dust depletion may also contribute.

These results demonstrate the potential of high-resolution tomography with gravitational arcs for studying the spatial, kinematic, and chemical structure of the CGM on small physical scales.

1. Introduction

Developing a complete picture of galaxy evolution requires understanding the role of the circumgalactic medium (CGM), which serves as the interface between galaxies and their surrounding environment (Tumlinson et al. 2017; Péroux & Howk 2020; Faucher-Giguère & Oh 2023). Over the past decades, absorption-line spectroscopy of background quasars has become one of the primary tools for probing the CGM, providing detailed constraints on its physical, chemical, and kinematic properties (Nielsen et al. 2013; Huang et al. 2021) . However, each quasar probes the halo along only a single pencil-beam sight-line, limiting our ability to investigate the spatial distribution, kinematic, and small-scale structure of the absorbing gas within individual halos.

A more direct view of the CGM can be obtained through multiple sight-lines intersecting the same halo, allowing spatial variations in absorber properties to be measured within individual galaxies (Steidel et al. 2010; Rubin et al. 2018). Strong gravitational lensing provides an ideal opportunity by producing multiple closely separated sight-lines through the same foreground halo, making it an ideal tool for tomographic studies of the CGM. (e.g., Lopez et al. 2018, 2020, 2024; Tejos et al. 2021; Mortensen et al. 2021; Fernández-Figueroa et al. 2022; Afruni et al. 2023; Shaban et al. 2025; Hernández-Guajardo et al. 2026).

Most existing tomographic studies have relied on low- or intermediate-resolution spectroscopy. Although these observations have revealed the large-scale distribution of CGM absorbers, they can not fully resolve their internal velocity structure. High-resolution spectroscopy of spatially extended background sources can provide access to the detailed kinematics, ionization structure, and small-scale variations of the absorbing gas while simultaneously preserving spatial information across multiple sight-lines. The feasibility and scientific potential of this approach were demonstrated by Diamond-Stanic et al. (2016), who used high-resolution spectroscopy of an extended background source to investigate intervening absorption systems at $z = 0.4$.

In this work, we use the unique geometry of the brightest known strongly lensed arc, the 'Sunburst Arc' (see Fig. 1), located behind the galaxy cluster PSZ1 G311.65–18.48 (Dahle et al. 2016). The arc is produced by strong gravitational lensing of a star-forming galaxy at $z = 2.369$ by a massive cluster at $z = 0.443$, providing multiple closely separated sight-lines through intervening absorbers. An intervening Mg II–Fe II absorption system at $z = 0.73379$ was identified by Lopez et al. (2020) toward the northernmost portion of the arc. Nebular [O II] emission detected at the same redshift reveals a nearby galaxy, which we identify as the absorber (hereafter 'G1'). Using this system, we aim to investigate the spatial distribution, kinematics, and chemical properties of low-ionization gas on kiloparsec scales.

2. Observation

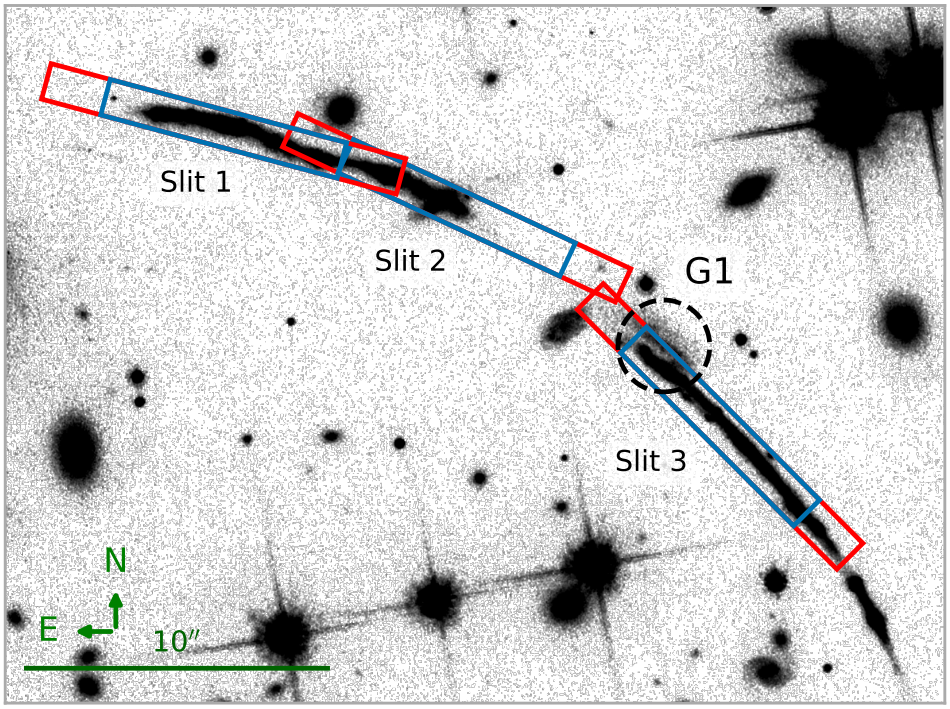


Fig. 1: Hubble Space Telescope (HST) image of the northernmost region of the 'Sunburst Arc' showing the three UVES slit used in this work. Slit 3 was aligned with the major axis of the foreground galaxy 'G1', highlighted by the black circle.

We observed three regions along the northernmost part of the 'Sunburst Arc' (Fig. 1) using the Ultraviolet and Visual Echelle Spectrograph (UVES; Dekker et al. 2000) mounted on the Very Large Telescope (VLT).

The observations were obtained as part of ESO Programme 113.26KJ.001 (PI: S. Lopez), with a total exposure time of ∼ 13 hr. The UVES slit dimensions were $1.2'' \times 8.0''$ in the blue arm and $1.2'' \times 12.0''$ in the red arm. Four individual exposures were obtained for each of slits 1 and 2, while slit 3 was observed in five exposures. The extra exposure of slit 3 was acquired as a repeat observation to satisfy the seeing constraint. All observations were carried out under dark-sky conditions with a median seeing of $0.73''$ on 2024 April 4 and 11, and May 2.

3. Data reduction and processing

The data were reduced using the `ESOReflex` pipeline (Freudling et al. 2013). Since the Sunburst Arc is an extended source, the pipeline's default sky subtraction was not applied. Instead, a master sky spectrum was constructed using uncontaminated regions of slit 2 (Fig. 1). After aligning the exposures onto a common wavelength grid with SpectRes (Carnall 2017), the sky spectra were combined using the median at each wavelength and smoothed with a Gaussian kernel. This master sky was subtracted from slit 2 and scaled to the lowest-flux regions of slits 1 and 3 before subtraction.

The spectra were converted to vacuum wavelengths and corrected to the barycentric reference frame. The final dataset comprises 14 spatial positions in the blue arm ($\Delta\lambda = 0.027$ Å pixel^{-1}; $0.57''$ per pixel) and 30 in the red arm ($\Delta\lambda = 0.029$ Å pixel^{-1}; $0.40''$ per pixel). The two-dimensional spectra were rebinned onto a uniform wavelength grid with a step of 0.04 Å, and one-dimensional spectra were extracted by computing the weighted average within each aperture (Fig. 2).

These processing steps slightly degrade the nominal UVES resolution ($R \sim 45\,000$). We estimated the final resolution by measuring the FWHM of unresolved sky emission lines in each arm (Fig. A.1), obtaining $R \sim 33\,000$ (9 km s^{-1}) at 4800 Å in the red arm and $R \sim 30\,000$ (10 km s^{-1}) at 4100 Å in the blue arm.

4. Absorption-line analysis

4.1. Pseudo-spaxel definition

Figure 2 shows the spatial configuration of the UVES blue and red arms. We defined eight spatial apertures (hereafter pseudo-spaxels), each approximately $1.2'' \times 1.2''$ in size, corresponding to 4×7 kpc after delensing. Each pseudo-spaxel combines three spatial positions in the red arm and two in the blue arm, except for Ap. 8, which has no blue-arm counterpart.

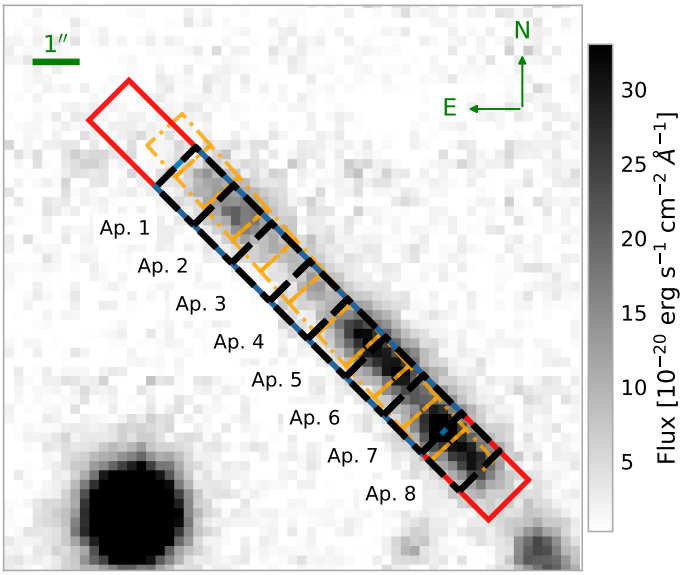


Fig. 2: Position of slit 3 on a MUSE narrow band image centered at ~ 4848 Å. The eight UVES apertures (black boxes), the eleven MagE apertures from Lopez et al. (2020) (orange boxes), and the UVES blue and red arms are shown.

4.2. Absorption-line measurements

For each pseudo-spaxel, we simultaneously fitted Voigt profiles to the Mg II $\lambda\lambda$2796, 2803 and Fe II $\lambda\lambda\lambda$2374, 2382, 2586 transitions, assuming a common redshift (z) and Doppler parameter (b) for all lines. The fits were performed using VoigtFit (Krogager 2018). We tested models with different numbers of velocity components and found that a single-component model provides the best description of the data in all pseudo-spaxels. The best-fitting parameters are summarized in Table 1, while the Mg II and Fe II fits are shown in Fig. 3 and Appendix B.1, respectively. For visualization, the spectra were rebinned to 0.2 Å.

Rest-frame equivalent widths (W_r) were calculated using the analytic approximation of Draine (2011) and the best-fitting Voigt profile parameters. The impact parameters (D), measured from the centers of the pseudo-spaxels to galaxy G1, were derived using the lens model of Lopez et al. (2020). The resulting W_r and D values are listed in Table 1.

We use the observed Mg II/Fe II column-density ratio as a proxy for the relative α-to-iron abundance, although this interpretation is subject to uncertainties from ionization and dust depletion. Moreover, since the Mg II transitions are strongly saturated in most apertures, the derived N(Mg II) values should be regarded as lower limits. Consequently, the inferred Mg II/Fe II ratios also provide lower limits on the intrinsic ratios.

No significant Mg II absorption is detected in slits 1 and 2, consistent with Lopez et al. (2020). The faintness of the extended source results in low signal-to-noise ratios (S/N), typically S/N $\approx$ 2 for Mg II and S/N $\approx$ 1 for Fe II. These values prevent us from deriving meaningful upper limits on the Mg II column densities; therefore, we do not report constraints for these slits.

Table 1: Summary of the absorption properties for each pseudo-spaxel, assuming a single component model. The columns list the pseudo-spaxel number; the impact parameter D from galaxy 'G1' (in kpc), the velocity relative to a z_{abs} = 0.73379 (in km s^{-1}) , the Doppler parameter b (in km s^{-1}), the column densities (log N in cm^{-2}) for Mg II and Fe II, their ratio log N(Mg II/Fe II), and the rest-frame equivalent widths (W_r) for selected transitions (in Å). Uncertainties correspond to 1σ errors.

Ap.	D [kpc]	v [km s^{-1}]	b [km s^{-1}]	log N [cm^{-2}]		log N(Mg II/Fe II)	W_r [Å]		
				Mg II	Fe II		Mg II 2796	Fe II 2382	Fe II 2586
1	3.4	$54.3^{+9.0}_{-9.8}$	$52.6^{+10.0}_{-14.9}$	$14.27^{+0.30}_{-0.06}$	$15.55^{+0.15}_{-0.05}$	$-1.28^{+0.30}_{-0.16}$	1.58 ± 0.30	1.11 ± 0.19	1.70 ± 0.34
2	4.9	$60.0^{+4.6}_{-3.0}$	$65.9^{+7.5}_{-5.3}$	$14.19^{+0.04}_{-0.08}$	$15.23^{+0.12}_{-0.10}$	$-1.03^{+0.11}_{-0.14}$	1.82 ± 0.14	1.22 ± 0.07	2.06 ± 0.25
3	8.1	$70.2^{+8.8}_{-8.6}$	$71.7^{+10.8}_{-15.6}$	$14.00^{+0.14}_{-0.13}$	$14.91^{+0.13}_{-0.12}$	$-0.91^{+0.19}_{-0.18}$	1.73 ± 0.22	1.04 ± 0.08	1.82 ± 0.25
4	12.0	$86.4^{+2.3}_{-2.6}$	$58.0^{+8.5}_{-5.5}$	$13.86^{+0.02}_{-0.04}$	$14.75^{+0.04}_{-0.07}$	$-0.89^{+0.07}_{-0.06}$	1.35 ± 0.11	0.79 ± 0.03	1.25 ± 0.10
5	15.9	$97.3^{+1.3}_{-4.5}$	$52.3^{+11.0}_{-15.2}$	$13.69^{+0.01}_{-0.02}$	$14.10^{+0.06}_{-0.02}$	$-0.41^{+0.02}_{-0.07}$	1.10 ± 0.17	0.57 ± 0.02	0.50 ± 0.01
6	19.9	$85.1^{+1.2}_{-1.8}$	$40.1^{+6.1}_{-3.1}$	$13.49^{+0.01}_{-0.01}$	$13.60^{+0.05}_{-0.03}$	$-0.11^{+0.03}_{-0.05}$	0.78 ± 0.05	0.38 ± 0.01	0.18 ± 0.01
7	24.1	$55.1^{+5.7}_{-14.9}$	$55.7^{+10.0}_{-17.6}$	$13.28^{+0.03}_{-0.21}$	$13.41^{+0.18}_{-0.08}$	$-0.13^{+0.09}_{-0.28}$	0.63 ± 0.00	0.27 ± 0.01	0.12 ± 0.01
8	28.3	$65.8^{+3.8}_{-1.6}$	$73.8^{+10.0}_{-17.6}$	$12.95^{+0.03}_{-0.03}$	—	—	0.35 ± 0.01	—	—

5. Results

Figure 3 shows the observed Mg II absorption profiles and their best-fitting models. The absorption is strongly saturated, and the profiles are remarkably similar across the different apertures. In most pseudo-spaxels (Ap. 1–6), the absorption reaches nearly zero

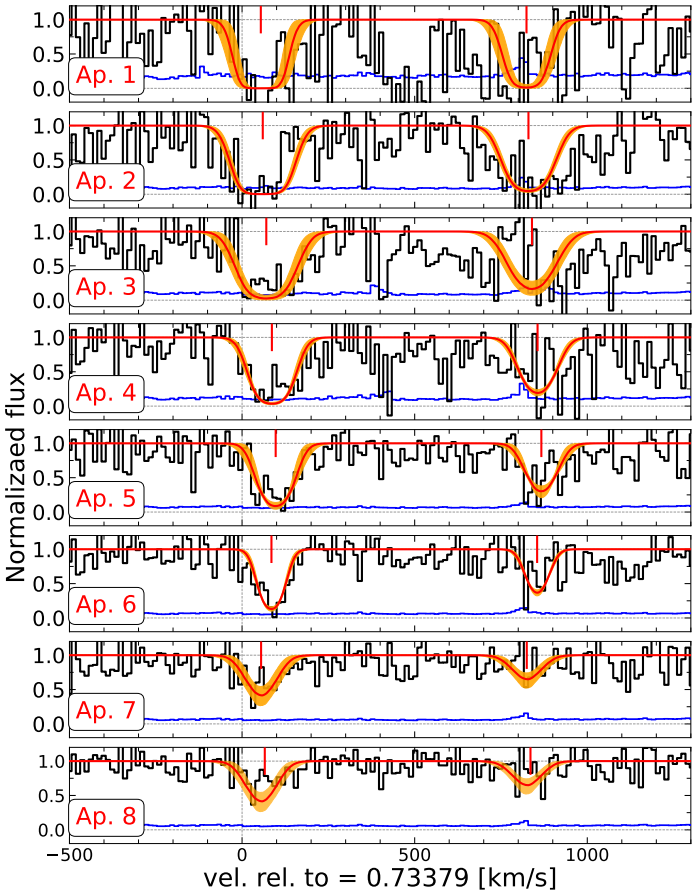


Fig. 3: Mg II absorption line fits to the eight UVES spectra for the $\lambda\lambda$2796, 2803 transitions. The black line represents the spectrum binned to 0.2 Å; the corresponding uncertainties are shown in blue. The red line shows the best-fitting Voigt profile, with the orange shaded region indicating the 1σ uncertainty of the fit. Red vertical ticks mark the velocity centroids of the fitted absorption components.

flux at line center, indicating a high covering fraction of the absorbing gas. Furthermore, no clear evidence of multiple resolved velocity components is found in any pseudo-spaxel.

The fitted Doppler parameters are systematically large, with a weighted mean of 58.8 ± 4.4 km s^{-1}. This is approximately ten times larger than the typical values measured for individual quasar absorption components (5.7 ± 4.4 km s^{-1}; Churchill et al. 2020). This suggests that the fitted profiles trace the effective velocity dispersion of multiple unresolved absorbing clouds within each pseudo-spaxel, rather than individual kinematic components.

As expected from the saturation of the Mg II transitions, the derived N(Mg II) values should be regarded as lower limits. Nevertheless, the observed log N(Mg II)/N(Fe II) ratio increases systematically with impact parameter (Table 1). This trend is qualitatively consistent with an increasing relative α-element abundance toward larger galactocentric distances along the galaxy's major axis, suggesting a possible chemical enrichment gradient in the CGM.

6. Discussion

Gravitational arcs provide multiple sight-lines through a single background source, offering an observational approach that is generally not possible with quasars. This allows us to probe the absorbing gas on galactic scales while also investigating its spatial distribution on smaller scales within individual pseudo-spaxels.

6.1. Impact of spatial averaging on column density measurement

To compare our pseudo-spaxel measurements with pencil-beam observations, we selected 49 high-resolution quasar sight-lines from Churchill et al. (2020), corresponding to seven comparison systems for each pseudo-spaxel. The sample was selected to have Mg II λ2796 equivalent widths comparable to those measured in our UVES pseudo-spaxels and to exhibit a single dominant absorption complex (see Fig. C.1). The quasar spectra were degraded to match the spectral resolution of our observations and fitted using the same single-component model adopted for the pseudo-spaxels.

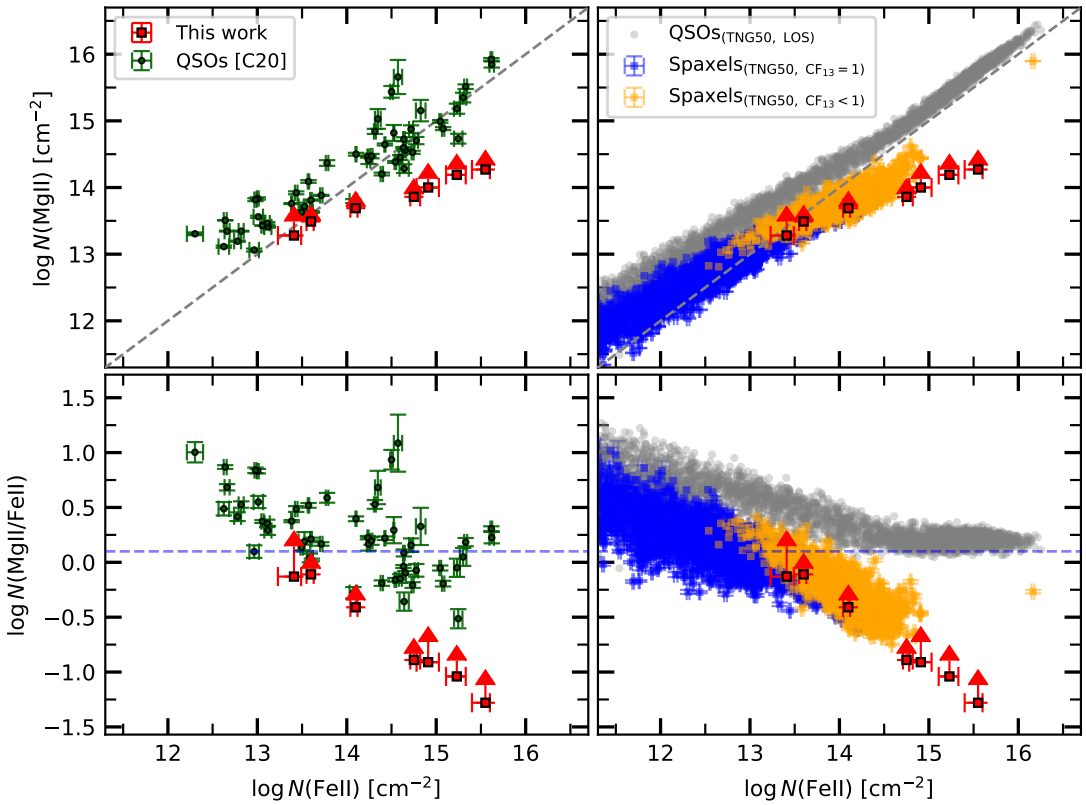


Fig. 4: Comparison of the Mg II and Fe II column densities (top panels) and the Mg II/Fe II column-density ratio as a function of Fe II column density (bottom panels). The left panels compare the UVES pseudo-spaxels from this work (red squares) with quasar absorption systems (green circles) selected to match the Mg II equivalent widths of the UVES pseudo-spaxels and fitted with a single absorption component. The right panels show the TNG50 results, including quasar sight-lines (gray circles) and pseudo-spaxels (squares). Orange and blue squares shows pseudo-spaxels with full (CF = 1) and partial (CF < 1) covering fractions, respectively. The blue dashed line marks the solar Mg II/Fe II ratio from Asplund et al. (2021).

As shown by the green squares in the left panels of Fig. 4, the quasar measurements follow similar overall trends to the UVES results but yield systematically higher column densities. This offset suggests that the pseudo-spaxel measurements are affected by spatial averaging over multiple sight-lines, an effect absent from pencil-beam quasar observations.

To investigate this effect, we performed synthetic observations using the TNG50 simulation (Nelson et al. 2019). We selected 198 isolated galaxies (subhalos) at $z = 1$, with stellar masses and star formation rates similar to those of galaxy G1 ($\log(M_\star/M_\odot) = 9.7 \pm 0.3$ and SFR $= 1.1 \pm 0.3$ $M_\odot, yr^{-1}$). For each galaxy, we generated synthetic quasar sight-lines extending to two virial radii using `Trident` and measured the CGM absorption. The resulting column densities are shown as grey circles in Fig. 4, where they follow a distribution similar to that of the degraded quasar sample.

To reproduce the UVES observations, we constructed synthetic slits matching the aperture sizes and projected distances of our observations. Each pseudo-spaxel was generated by combining the spectra of 28 individual sight-lines, followed by spectral degradation and rebinning to match the UVES data.

Since the individual sight-lines within a pseudo-spaxel do not necessarily intersect the same absorbing structures, some may show weak or no detectable absorption. We therefore classified the simulated pseudo-spaxels according to their Mg II covering fraction. We defined CF = 1 for pseudo-spaxels in which all 28 sight-lines have $N(\mathrm{MgII}) > 10^{13}$ cm^{-2}, and CF < 1 when at least one sight-line falls below this threshold. The resulting synthetic spectra were then fitted using the same single-component model adopted for the UVES pseudo-spaxels and quasar sight-lines.

The recovered column densities are shown in the right panels of Fig. 4. Orange and blue circles represent pseudo-spaxels with CF = 1 and CF < 1, respectively. Pseudo-spaxels with incomplete covering fractions rarely reproduce the observed UVES column densities, whereas those with CF $\simeq$ 1 provide a better match to the measured Mg II and Fe II column densities and their ratios. This suggests that the observed absorption is most consistent with gas covering nearly the entire area sampled by each pseudo-spaxel.

Overall, these comparisons indicate that the recovered UVES column densities are affected by both line saturation and spatial averaging. The strong saturation of the Mg II transitions leads to lower limits on the intrinsic column densities. In addition, combining multiple neighboring sight-lines into a single pseudo-spaxel modifies the absorption profiles and can further affect the column densities recovered from single-component fits. The better agreement between the UVES measurements and simulated pseudo-spaxels with CF $\simeq$ 1 suggests that spatial averaging plays an important role in explaining the observed offset relative to pencil-beam measurements. It also indicates that the absorbing gas is spatially extended over the kiloparsec scales probed by our observations.

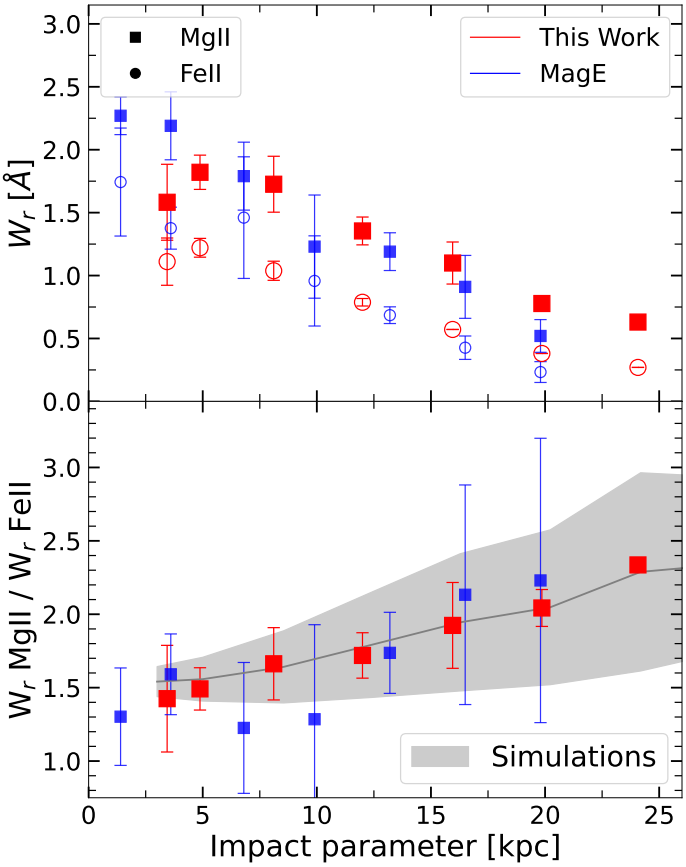


Fig. 5: Comparison of the rest-frame equivalent widths of Mg II λ2796 and Fe II λ2382, and their ratio, as a function of impact parameter. Red and blue symbols represent this work and the MagE measurements from Lopez et al. (2020), respectively. Top: Mg II λ2796 (squares) and Fe II λ2382 (circles). Bottom: The Mg II/Fe II equivalent-width ratio, used as a proxy for relative α-to-iron enrichment. The increasing ratio toward larger galactocentric distances suggests a chemical gradient, with more α-enriched material in the inner CGM.

The column densities presented in the previous section recover the overall trends in the absorption properties but should be regarded as lower limits due to line saturation. To assess whether these trends persist using an observable less affected by the uncertainties in our low S/N data, we analyze the rest-frame equivalent widths. Figure 5 shows the equivalent widths of Mg II and Fe II, together with their ratio, as a function of impact parameter.

The equivalent widths of both Mg II and Fe II decrease with increasing projected distance from galaxy G1, indicating weaker absorption at larger galactocentric distances. More importantly, the ratio W_r(Fe II)/W_r(Mg II) also decreases systematically with increasing $D/R_{\rm vir}$. This trend is consistent with the results of Lopez et al. (2020) based on MagE observations, suggesting a relative increase in α-element enrichment toward the inner CGM.

To further investigate this trend, we compare the observations with the simulated pseudo-spaxels having CF = 1. The grey shaded region in the bottom panel of Fig. 5 represents the mean ratio and its 1σ dispersion from these simulations. The simulated trend is consistent with the observed measurements, providing complementary support for a chemical enrichment gradient in the CGM, with relatively more α-enhanced material toward the inner regions.

Several physical mechanisms may contribute to the observed radial variation in the Mg II/Fe II ratio, including ionization, dust depletion, and chemical enrichment. From the perspective of photoionization, Mg II and Fe II have similar first ionization potentials (15.04 and 16.19 eV, respectively), and therefore coexist over a broad range of weakly ionized gas conditions. Although iron is more readily ionized to Fe III than magnesium is to Mg III, ionization alone is unlikely to produce the coherent radial trend observed over tens of kiloparsecs. Dust depletion provides another possible explanation. Iron is a refractory element and is more efficiently incorporated into dust grains than magnesium. Consequently, if the dust content decreases with galactocentric distance, the observed increase in the Mg II/Fe II ratio could arise naturally from a reduction in iron depletion. Quantifying this effect would require applying depletion corrections (e.g., De Cia et al. 2016), which is beyond the scope of the present work because the available data do not provide sufficient constraints on the depletion strength and intrinsic abundance pattern. An alternative explanation is a gradient in chemical enrichment. Because α elements are predominantly produced by core-collapse supernovae whereas iron is progressively enriched by Type Ia supernovae, an increasing Mg II/Fe II ratio with galactocentric distance is consistent with the outer CGM being

more α-enhanced (or equivalently less Fe-enriched) than the inner regions. This interpretation is also consistent with the results of Lopez et al. (2020), who argued that the outer CGM surrounding G1 is less chemically evolved than the gas closer to the galaxy.

7. Conclusions

We have used high-resolution spectroscopy of the 'Sunburst Arc' to study an intervening Mg II–Fe II absorption system at $z = 0.73379$, using multiple closely spaced sight-lines through the CGM of the absorber galaxy 'G1'. This unique configuration allows us to probe the spatial structure, kinematics, and chemical properties of low-ionization circumgalactic gas on scales that are inaccessible with single quasar sight-lines.

1. Strong Mg II absorption is detected across all pseudo-spaxels, extending to projected distances of ~ 28 kpc from G1. This indicates a covering fraction close to unity over the scales sampled by our observations. The absorption profiles remain remarkably similar across the lensed arc, suggesting that the low-ionization CGM is distributed over extended regions rather than confined to isolated compact clouds.
2. Despite the high spectral resolution of the UVES observations, the absorption profiles in all pseudo-spaxels are well described by a single-component model. The measured Doppler parameters ($b = 58.8 \pm 4.4$ km s^{-1}) are approximately an order of magnitude larger than those typically measured for individual quasar absorption components. This suggests that the observed line widths trace the combined velocity dispersion of multiple unresolved absorbing clouds within each pseudo-spaxel, rather than the kinematics of a single physical cloud.
3. Comparisons with matched quasar sight-lines and synthetic TNG50 observations indicate that the recovered Mg II column densities are affected by both line saturation and spatial averaging within the pseudo-spaxels. Nevertheless, the systematic increase in the Mg II/Fe II ratio with galactocentric distance is seen in both the column densities and equivalent widths, suggesting a radial variation in the physical and chemical properties of the CGM. Although ionization and dust depletion may also contribute to this trend, it is qualitatively consistent with increasing relative α-element enrichment toward larger galactocentric distances.

These results demonstrate the potential of high-resolution tomography with gravitational arcs to probe the spatial, kinematic, and chemical structure of the circumgalactic medium. They also highlight the effects of spatial averaging across extended sources, which must be considered when interpreting absorption-line measurements.

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Appendix A: Resolution

Appendix B: Fe II absorption-line fits

Appendix C: Quasar sight-line comparison

Appendix D: rEW comparison

Figure D.1 also compares our measurements with two other studies based on gravitationally lensed arcs. Fernández-Figueroa et al. (2022); Tejos et al. (2021) identified Mg II absorption along both the major and minor axes of galaxies, while Hernández-Guajardo et al. (2026) studied Mg II absorption associated with minor-axis outflows. In all cases, we observe the same general trend of decreasing Mg II equivalent width with increasing impact parameter. For the W_r(Fe II)/W_r(Mg II) ratio, however, a comparison with the major-axis measurements from Fernández-Figueroa et al. (2022); Tejos et al. (2021) is not possible because the Fe II transitions are not covered by the MUSE wavelength range. We can instead compare our measurements with the minor axis measurements of Hernández-Guajardo et al. (2026), which show a broadly consistent behavior. Given the limited number of measurements, however, we cannot rule out a different behavior of the Mg II/Fe II ratio between the major and minor axes at this redshift. A larger sample of individual halos with sight-lines aligned with both orientations is required to robustly test for such an orientation dependence.

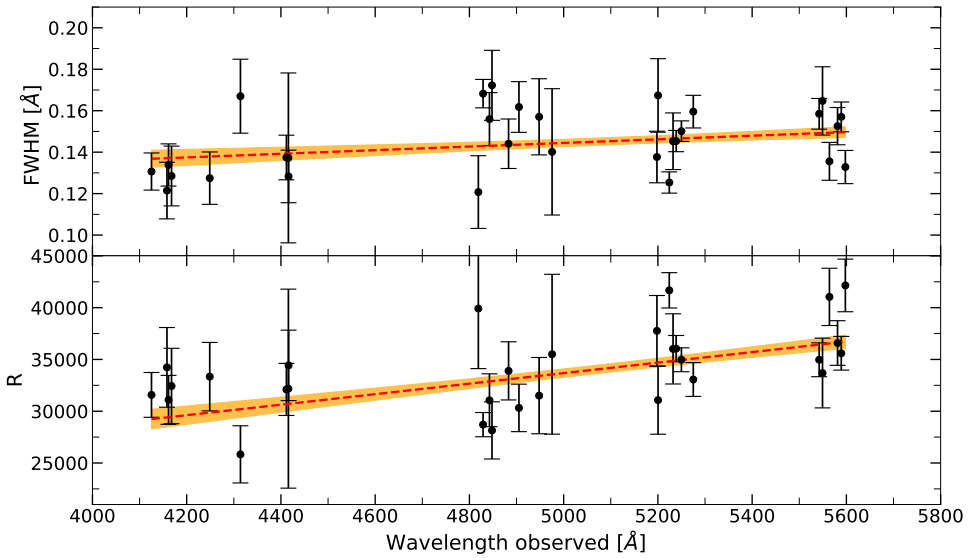


Fig. A.1: FWHM and resolution for slit 3 from sky emission line from the combined spectrum.

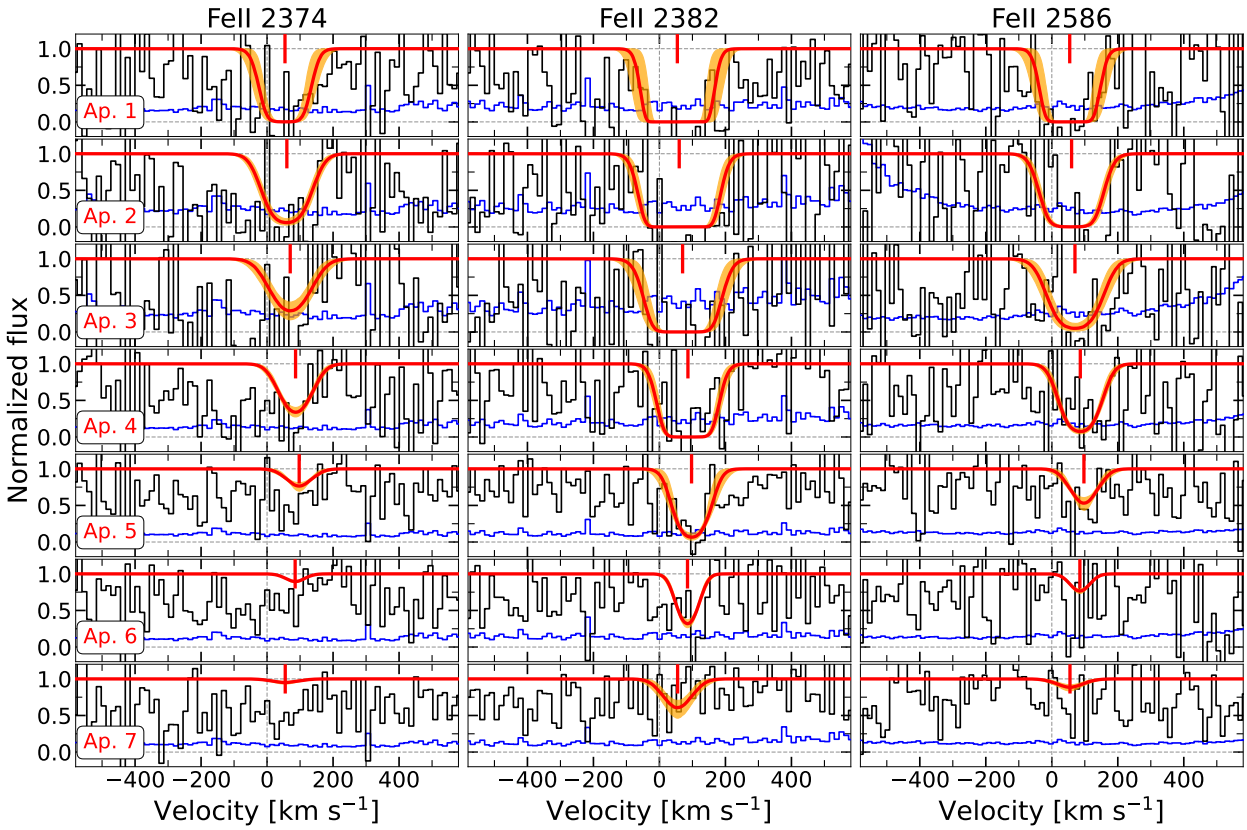


Fig. B.1: Absorption lines and fits of Fe II $\lambda\lambda$2374, 2382, 2586.

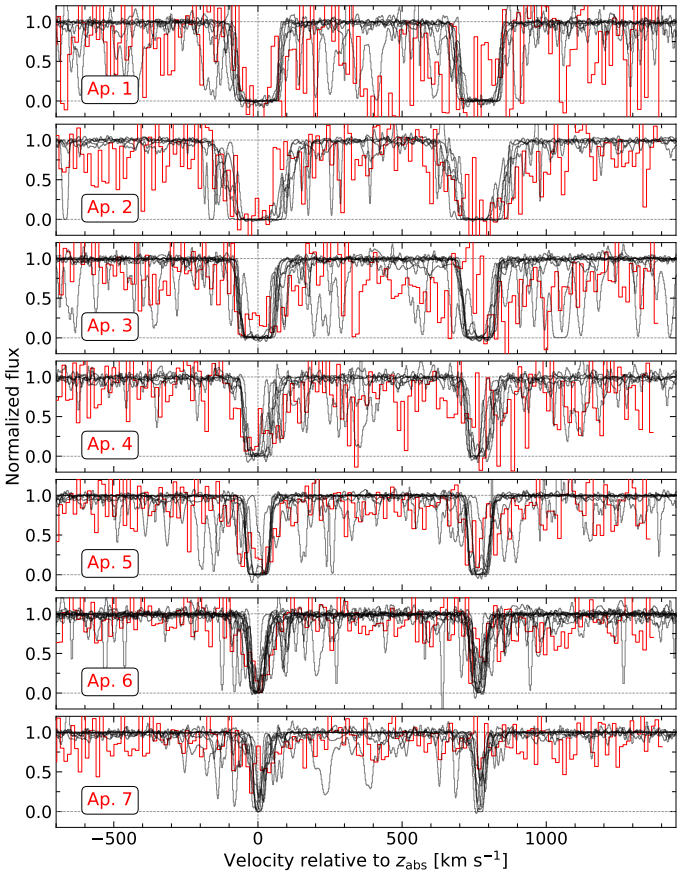


Fig. C.1: Comparison between apertures 1–7 (red) and a sample of ∼ 7 quasar sight lines (grey), with the quasar spectra degraded to match the UVES spectral resolution. This comparison highlights the similarity between the absorption properties observed along the lensed galaxy sight lines and those seen toward background quasars.

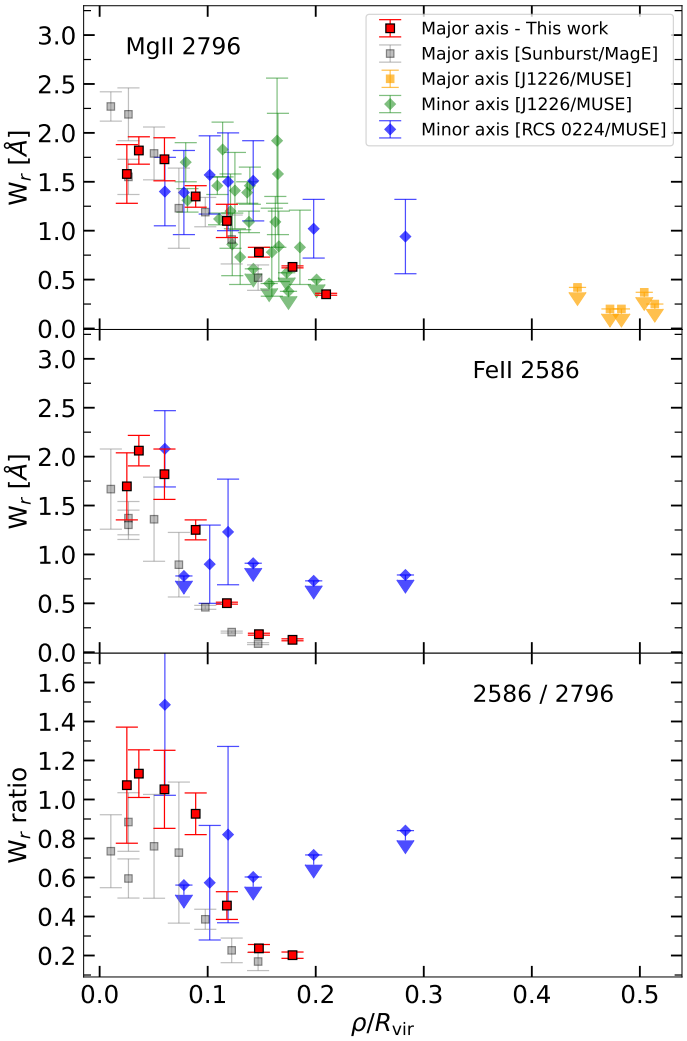


Fig. D.1: Comparison of the rest-frame equivalent widths of Fe II λ2586 and Mg II λ2796, and their ratio, as a function of impact parameter normalized by the virial radius (b/R_{vir} = 135 kpc; Lopez et al. 2020). Red squares show the UVES measurements from this work. We compare our measurements with two studies of gravitationally lensed arcs observed with MUSE: RCS 0224 from Hernández-Guajardo et al. (2026), which probes Mg II absorption along the minor axis, and J1226 from Fernández-Figueroa et al. (2022); Tejos et al. (2021), which provides measurements along the minor axis and upper limits along the major axis.